

QUA'TARIOUS A. BOWMAN

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Portfolio: [resume-pink-three.vercel.app](#)

[QR CODE]

TECHNICAL ARSENAL

CAD & MODELING

- SolidWorks
- Fusion 360
- Parametric Design
- Mechanical Assemblies

FABRICATION & TOOLS

- FDM 3D Printing
- Micro-Scale Soldering (IPC Hand Soldering Standards)
- Structural Carpentry
- Mixed-Material Integration
- Workshop Tooling

HARDWARE SYSTEMS

- LEV Commuter Integration
- High-Current Wire Harnesses
- FPV Drone Electronics

RAPID SOFTWARE

- React & Tailwind CSS
- Python (Automation)
- Git Version Control

EDUCATION

University of Virginia (UVA)

B.S. Mechanical Engineering

Expected 2028

ENGINEERING PORTFOLIO

Potential Energy Propulsion Vehicle

Lead CAD & Prototyping

Spring 2026

- Co-designed and manufactured a dual-propulsion mechanism featuring a 20.62g ABS chassis, a two-spring potential energy drivetrain, and a 2.5:1 ratio gear train.
- Executed rapid design iterations and assembly layouts in CAD to optimize torque metrics and structural stability.

Custom Light Electric Vehicle (LEV)

Systems Builder

Summer 2026

- Integrated a 1000W rear-wheel drive e-bike system, engineering custom mounting hardware and troubleshooting mechanical alignment.
- Identified and resolved factory framing flaws to ensure structural integrity under high-load commuter conditions.

5-Inch FPV Drone Assembly

Hardware Tech

Spring 2026

- Executed micro-scale precision soldering to build a high-density electronics stack, including a 4-in-1 Electronic Speed Controller (ESC).
- Configured custom power and data wiring harnesses, ensuring clean physical routing and vibration-resistant mounting.

Immersive VR Workstation

Integration Designer

Summer 2026

- Engineered a suspended structural cable harness and continuous charging system to optimize physical ergonomics for prolonged VR productivity.

Experimental Field Testing

Engineering Lab

Spring 2026

- Developed and executed structural quality testing guidelines for physical prototypes, establishing baseline metrics for material durability.